

ws 1125 Let $A = PJP^{-1}$, J JNF

$$\textcircled{1} \operatorname{tr} A = \sum_{i=1}^n \lambda_i$$

$$\operatorname{tr}(A) = \operatorname{tr}(PJP^{-1}) = \operatorname{tr}(\cancel{P^{-1}P}J) = \operatorname{tr} J = \sum_{i=1}^n \lambda_i$$

$$\textcircled{2} \det(A) = \prod_{i=1}^n \lambda_i \quad (\det(P^{-1}) = 1/\det(P))$$

$$\begin{aligned} \det(A) &= \det(PJP^{-1}) = \cancel{\det(P)} \det(J) \cancel{\det(P^{-1})} \\ &= \prod_{i=1}^n \lambda_i \end{aligned}$$

$\uparrow$
upper triangular

Cayley-Hamilton Theorem

$A \in M_n(\mathbb{C})$, w/ characteristic polynomial $\chi_A: \mathbb{C} \rightarrow \mathbb{C}$

$$\begin{aligned} \chi_A(\lambda) &= p_A(\lambda) = \det(\lambda I_n - A) \\ &= \lambda^n + c_{n-1}\lambda^{n-1} + c_{n-2}\lambda^{n-2} + \dots + c_1\lambda + c_0 \end{aligned}$$

then $\chi_A(A) = 0$ for $\chi_A: M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$

$$\text{w/ } \chi_A(z) = z^n + c_{n-1}z^{n-1} + \dots + c_1z + c_0 I_n$$

$$\text{Ex } A = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}, \quad \chi_A(\lambda) = \lambda^2 - 6\lambda + 5$$

$$\chi_A(A) = A^2 - 6A + 5I_2$$

$$= \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}^2 - 6 \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix} + 5 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \dots = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} =: 0 \in M_2(\mathbb{R})$$

Pf Suppose $A = P J P^{-1}$ as Jordan Decomposition

s.t., $J = \text{diag}(J_1, \dots, J_k)$ w/ J_i of size $m_i \times m_i$

e.g., $J = \begin{bmatrix} \boxed{2} & & & & \\ & \boxed{2} & & & \\ & & \begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix} & & \\ & & & \begin{bmatrix} 3 \\ 3 \end{bmatrix} & \\ & & & & \boxed{3} \end{bmatrix}$

$m_1 = 1$
 $m_2 = 1$
 $m_3 = 2$
 $m_4 = 1$
 $m_5 = 1$

$$\chi_A(\lambda) = (\lambda - 2)(\lambda - 2)(\lambda - 2)^2(\lambda - 3)(\lambda - 3)$$

$$\lambda_1 = \lambda_2 = \lambda_3 = 2, \quad \lambda_4 = \lambda_5 = 3$$

then the characteristic polynomial can be written as

$$\chi_A(\lambda) = (\lambda - \lambda_1)^{m_1} (\lambda - \lambda_2)^{m_2} \dots (\lambda - \lambda_k)^{m_k}$$

where λ_i 's being eigenvalues of A (not necessarily distinct)

Note that $\chi_A(A) = P \chi_A(J) P^{-1}$

it suffices to verify that $\chi_A(J) = 0$

Then since

$$\chi_A(J) = (J - \lambda_1 I_n)^{m_1} (J - \lambda_2 I_n)^{m_2} \dots (J - \lambda_k)^{m_k}$$

but all factors are block diagonal

So it suffices to show that for any fixed eigenvalue λ_i w/ the largest Jordan block (among the same eigenvalue) that

$$(J_i - \lambda_i I_{m_i})^{m_i} = 0$$

which is indeed true. □

e.g. $J_i = \begin{pmatrix} \lambda_i & 1 & & \\ & \lambda_i & 1 & \\ & & \lambda_i & 1 \\ & & & \lambda_i \end{pmatrix}_{4 \times 4}$ is the largest Jordan block corresponding to the e-val λ_i

$$J_i - \lambda_i I_4 = \begin{pmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & 0 & 1 \\ & & & 0 \end{pmatrix}$$

$$(J_i - \lambda_i I_4)^2 = \begin{pmatrix} 0 & 0 & 1 & \\ & 0 & 0 & 1 \\ & & 0 & 0 \\ & & & 0 \end{pmatrix}$$

$$(J_i - \lambda_i I_4)^3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ & 0 & 0 & 0 \\ & & 0 & 0 \\ & & & 0 \end{pmatrix}$$

$$(J_i - \lambda_i I_4)^4 = \begin{pmatrix} \bigcirc \end{pmatrix}$$

Rem Suppose A has distinct eigenvalues $\lambda_1, \dots, \lambda_m$

A is diagonalizable $\Leftrightarrow$

the polynomial p of lowest degree s.t.

$p(A) = 0$ is of the form

$$p(\lambda) = (\lambda - \lambda_1)^1 \dots (\lambda - \lambda_m)^1$$

(This polynomial is called the minimal polynomial of A , & $p(\lambda)$ divides $\chi_A(\lambda)$)

Pf obs that

$$\begin{pmatrix} x^2 + 5x & 2x + 1 \\ 3x & 6 \end{pmatrix} \in M_{2 \times 2}(\mathbb{C}[x]) = \mathbb{C}[x]^{2 \times 2}$$

is a matrix w/ polynomial
entries

$$x^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + x \begin{pmatrix} 5 & 2 \\ 3 & 0 \end{pmatrix} + 1 \begin{pmatrix} 0 & 1 \\ 0 & 6 \end{pmatrix} \in M_2(\mathbb{C})[x] = \mathbb{C}^{2 \times 2}[x]$$

is a polynomial w/ matrix coefficients,

$$A = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$$

$$\det(\lambda I - A) = \det \begin{pmatrix} \lambda - 4 & -1 \\ -3 & \lambda - 2 \end{pmatrix}$$

$$\left(\begin{array}{l} \det: \langle \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \rangle^{2 \times 2} \rightarrow \langle \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \rangle \\ \det \left(\begin{pmatrix} 4 \\ 3 \end{pmatrix} - 4 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad -1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ -3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 4 \\ 3 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) \\ = \begin{pmatrix} 4 \\ 3 \end{pmatrix}^2 - 6 \begin{pmatrix} 4 \\ 3 \end{pmatrix} + 5 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{array} \right)$$

$$(\lambda I_n - A)^{-1} = \frac{\text{adj}(\lambda I_n - A)}{\det(\lambda I_n - A)}$$

$$\Leftrightarrow \underbrace{(\lambda I_n - A)}_{\in M_n(\mathbb{C}[\lambda])} \underbrace{\text{adj}(\lambda I_n - A)}_{\substack{\in M_n(\mathbb{C}[\lambda]) \\ \text{matrix} \\ \text{multiplication}}} = \underbrace{\det(\lambda I_n - A)}_{\det: \mathbb{C}^{n \times n} \rightarrow \mathbb{C}} \underbrace{I_n}_{\text{adj}: \mathbb{C}^{n \times n} \rightarrow \mathbb{C}^{n \times n}}$$

$$\text{or } \underbrace{(\lambda I_n - A)}_{\in M_n(\mathbb{C})[\lambda]} \cdot \underbrace{\text{adj}(\lambda I_n - A)}_{\substack{\in M_n(\mathbb{C})[\lambda] \\ \text{mult. of poly.}}} = \underbrace{\det(\lambda I_n - A)}_{\text{poly. in } \lambda} I_n$$

□

$$\underline{E_x} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -3 & 2 & 4 \end{bmatrix}$$

$$\det(\lambda I_3 - A) = (\lambda - 2)^3 = 0 \Rightarrow \lambda = 2 \text{ w/ alg. mult.} = 3$$

$$\text{Let's solve } A_1 v = \lambda v_1, \text{ i.e., } (A - \lambda I_3)v = 0$$

$$\Rightarrow v_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \Rightarrow \text{geo. mult. of } \lambda = 2 \text{ is } = 1$$

Look for a **generalized eigenvector** v_2 s.t.

$$A v_2 = 2 v_2 + 1 v_1$$

$$\Rightarrow (A - 2I_3)v_2 = v_1$$

$$\text{solve and pick } v_2 = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$$

Look for **generalized eigenvector** v_3 s.t.

$$A v_3 = 2 v_3 + 1 v_2$$

$$\Rightarrow (A - 2I_3)v_3 = v_2$$

$$\text{solve & choose } v_3 = \begin{bmatrix} -2 \\ -3 \\ 0 \end{bmatrix}$$

so

$$A \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} | & | & | \\ A v_1 & A v_2 & A v_3 \\ | & | & | \end{bmatrix}$$

$$= \begin{bmatrix} | & | & | \\ 2v_1 & 2v_2 + v_1 & 2v_3 + v_2 \\ | & | & | \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ v_1 & v_2 & v_3 \\ 1 & 1 & 1 \end{bmatrix}}_P \underbrace{\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}}_J$$

FACT If A is symmetric, then A does not need generalized eigenvectors. (hence diagonalizable)

Otherwise for some eigenvalue λ ,

$\exists$ generalized eigenvector v s.t.

$$(A - \lambda I)^2 v = 0 \quad \& \quad (A - \lambda I)v \neq 0$$

(e.g. $Av_2 = \lambda v_2 + v_1$)
 $v_1 \neq 0$

$$\Rightarrow v^T (A - \lambda I)^2 v = 0$$

$$\text{but } v^T (A - \lambda I)^2 v = v^T \underbrace{(A - \lambda I)^T}_{A=A^T} (A - \lambda I) v$$

$$= \| (A - \lambda I)v \|^2 \neq 0$$
